www.tntextbooks.in

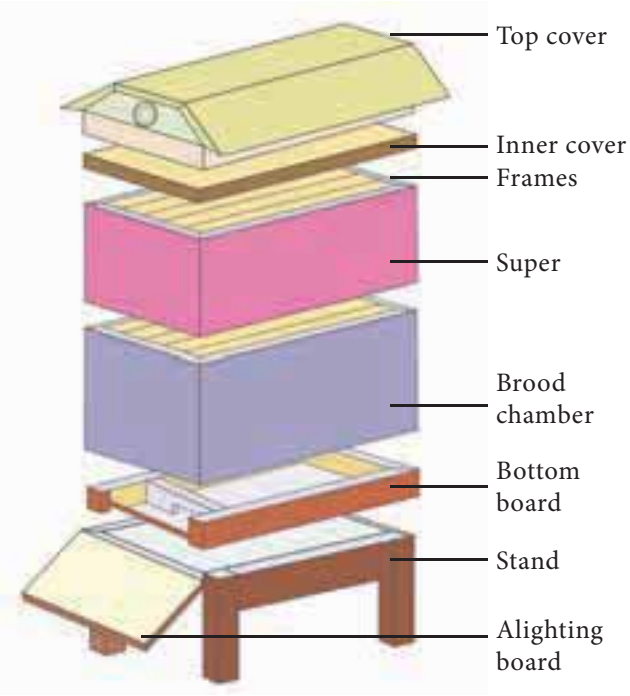

Figure 13.6 - Langstroth bee hive

bee from the brood chamber into the super chamber. **Comb foundation** is a sheet of bee wax, on both sides of which the exact shape of different cells of the comb is made in advance. **Bee gloves** are used by bee keepers for protecting their hands while inspecting the hives. **Bee veil** is a device made of fine nettings to protect the bee-keeper from bee sting. **Smoker** is used to scare the bees during hive maintenance and honey collection by releasing smoke. **Hive Tool** is a flat, narrow and long piece of iron which helps in scraping excess propolis or wax from hive parts. **Uncapping knife** is a long knife which helps in removing the cap from the combs as a first step in honey extraction. **Bee brush** is a large brush often employed to brush off bees from honey combs particularly at the time of extraction. **Queen introducing cage** is a pipe made of wire nets used for keeping the queen for about 24 hours for acquaintance with the hive and worker bees. **Feeder** is a basin with sugar syrup covered by grass to feed the bees during drought season. The grass prevents the bees from sinking into the syrup. **Honey Extractor** is a stainless-steel device which spins the combs rapidly to extract honey. **Hive Entrance Guard** is a device similar to queen excluder in front of the hive entrance which prevents the escape of queen during warming season.

Products of bee keeping and their economic importance

The chief products of bee keeping industry are honey and bee wax.

Honey is the healthier substitute for sugar. The major constituents of honey are: levulose, dextrose, maltose, other sugars, enzymes, pigments, ash and water. It is an aromatic sweet material derived from nectar of plants. It is a natural food, the smell and taste depends upon the pollen taken by the honey bee. It is used as an antiseptic, laxative and as a sedative. It is generally used in Ayurvedic and Unani systems of medicine. It is also used in the preparation of cakes, breads and biscuits

Bee wax is secreted by the abdomen of the worker bees at the age of two weeks. The wax is masticated and mixed with the secretions of the cephalic glands to convert it into a plastic resinous substance. The resinous chemical substance present in the wax is called **propolis** which is derived from pollen grains. The pure wax is white in colour and the yellow colour is due to the presence of carotenoid pigments. It is used for making candles, water proofing materials, polishes for floors, furniture, appliances, leather and taps. It is also used for the production of